

AI Crime Detection & Surveillance System

Problem Statement

Crime detection and surveillance management have become major challenges in modern cities and public environments because of the rapid increase in criminal activities, security threats, and public safety concerns. Security personnel often spend long hours continuously monitoring CCTV cameras and surveillance systems, which may lead to fatigue, delayed responses, and human errors during critical situations. Traditional surveillance systems mainly depend on manual observation and recording of surveillance footage instead of providing intelligent real-time crime analysis and automatic threat detection.

As a result, incidents such as suspicious activities, violence, robbery, trespassing, weapon usage, abnormal behavior, and dangerous movements may go unnoticed because of continuous monitoring and the huge amount of surveillance data generated every day. Another major challenge is the difficulty in integrating and analyzing different types of multimedia data such as surveillance videos, images, audio signals, and crime-related textual information together in a single intelligent system.

The proposed AI Crime Detection & Surveillance System uses multimodal Artificial Intelligence techniques to automatically analyze surveillance videos, images, audio data, and crime-related text information. The system can detect suspicious activities, recognize violence or weapon-related incidents, identify abnormal sounds such as alarms, screams, or gunshots, and support intelligent crime analysis using machine learning and deep learning techniques. The project aims to improve public safety, reduce manual monitoring effort, provide intelligent real-time surveillance support, enhance security management efficiency, and support faster emergency response in public places such as schools, shopping malls, banks, airports, hospitals, railway stations, parking areas, offices, and smart city environments.

Objectives

- To develop an intelligent AI-based surveillance monitoring system for automated crime detection.
- To detect suspicious activities and abnormal behavior using surveillance video analysis.
- To identify violence, robbery, and weapon-related incidents using image and video processing techniques.
- To recognize abnormal sounds such as alarms, screams, sirens, and gunshots using audio analysis.
- To analyze crime-related reports and textual information using Natural Language Processing techniques.
- To reduce human monitoring effort and improve surveillance management efficiency.
- To generate real-time alerts and support faster emergency response during dangerous situations.
- To improve public safety and support smart surveillance infrastructure in modern environments.